Challenge: Ramp Climber

Points: 200 • Estimated Time: 90 Minutes

Challenge Scope

The robot must climb an inclined wooden board set at a 15-degree angle and hold its position at the top.

Instructions & Engineering Hints

• **Objective:** Design and program a robot capable of successfully driving up a wooden ramp set at a 15-degree incline, reaching the top platform, and securely holding its position without slipping backward.

• **Structural Considerations:** Think about traction and weight distribution. Standard smooth plastic wheels might slip on wood; consider how you can increase friction using rubber bands, treads, or custom mechanical grippers. Also, placing heavier components strategically can help keep the drive wheels firmly pressed against the ramp surface.

• **Climbing Mechanics:** Standard gear ratios might stall under the load of gravity. Experiment with torque-prioritized gear trains to give your motors the extra strength needed to push the robot upward.

• **Holding Position:** Once the robot reaches the top, gravity will try to pull it back down. Think about whether your code needs to continuously apply motor power, or if a mechanical mechanism like a ratchet, worm gear, or brake system can passively hold the position.

• **Programming Strategy:** Utilize sensor feedback—such as a touch sensor bumper at the front or a gyroscope/tilt sensor—to detect when the robot has reached the flat surface at the top of the ramp so it knows when to stop and hold.